

Department of Information Technology

Central University of Kashmir

B.Tech. CSE

Semester VI					
S.No.	Course Code	Course Title	Credits	CIA	ESE
1	BTCS 609	Software Analysis & Design	2	50	50
2	BTCS 602	Computer Networks	4	50	50
3	BTCS 604	Theory of Computation	4	50	50
4	BTCS 606 L	Computer Networks Lab	2	50	50
5	BTCS 607	Microprocessor	4	50	50
6	BTCS 608 L	Microprocessor Lab	2	50	50
7	BTCS E**	Elective 1	4	50	50
		Total	22		

Central University of Kashmir

Course Title	Software Analysis and Design	Course Code	BTCS 609	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	02	L	T	P
Semester	6th	2	0	0

Course Objectives

- To understand the components and characteristics of a system, the types of information systems, and the role of a system analyst in system development.
- To understand the System Development Life Cycle (SDLC), SDLC models, and the different phases of the development process.
- To understand how to use various structured analysis tools to analyze and document the requirements of a system, how to conduct cost/benefit analysis to evaluate the feasibility of a project, and the different types of costs associated with system development.
- To understand the principles of modular and structured design, including defining module specifications, and understanding the concepts of coupling and cohesion.
- To understand the principles and techniques of system testing, including the development of a test plan, different testing techniques, quality assurance goals throughout the system life cycle, and current trends in testing.

Learning Outcomes

- Apply the knowledge to choose the appropriate system development methodology, including system development life cycle (SDLC) models, based on the requirements and constraints of the project.
- Apply techniques learned to ensure the system operates smoothly, meets users' needs, and remains up to date with evolving technology.
- Apply the knowledge to Plan and investigate a system development project by identifying its scope, goals, and objectives, to Conduct fact analysis and use various information gathering tools to collect relevant data for the project.
- Apply object-oriented analysis (OOA) and use Unified Modeling Language (UML) to model the system, including use cases, activity diagrams, class diagrams, and sequence diagrams.
- Apply various testing techniques to ensure the system is reliable, efficient, and meets user requirements and carry out software maintenance to fix bugs, improve performance, and add new features.

Course Synopsis

- Students will understand system planning, fact analysis, information gathering tools, structured analysis tools such as data flow diagrams and data dictionaries, cost/benefit analysis, system requirement specifications, and object-oriented analysis and modeling.
- Students will understand system testing and quality assurance, implementation and software maintenance, system control and audit trails, system administration and training, hardware/software selection, and project scheduling and management.
- Overall, the knowledge gained from these topics can be applied to analyze, design, develop, test, implement, and maintain information systems effectively and efficiently.

Course Outline / Content

Unit	Topics	Weeks
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1.	System Concepts and SDLC: Components and Characteristics of a System; Types of Information Systems; Modern Approach to System Analysis and Development, Role and Need of System Analyst in System Development; System Development Methodology: System Development Life Cycle, SDLC Models, Requirement Analysis, Feasibility Study and types, System Analysis and system Design, Link Program Testing, Conversion And Installation, System Review And Evaluation, Maintenance; Prototyping.	4
2.	System Analysis: System Planning and Initial Investigation, Fact Analysis, Information Gathering Tools; Tools of Structured Analysis: Data flow Diagram(DFD), Data Dictionaries, Decision Trees And Tables; Cost/Benefit Analysis, Types of costs ; System Requirement Specifications, Documentation Techniques for System Analysis; Object Oriented Analysis UML(Unified Modelling language), Object Oriented Development Life Cycle and Modelling, Modelling using UML (Use cases, Activity Diagram, Class Diagram, Sequence Diagram)	4
3.	System Design: Modular and Structured Design, Module Specifications, Coupling and Cohesion; Forms-Driven Methodology IPO Charts, Structured Walkthrough; Input/ Output and Forms Design: Requirements of Forms Design, Types of Forms; Dialog (User Interface) Design; File and Data Base Design: File Structure and File Organization, Data Structure, Normalization and its Types, Role of Data Base Administrator.	3
4.	System Implementation: System Testing and Quality Assurance, Test Plan, Testing Techniques Available, Quality Assurance Goals in Systems	4

	Life cycle, Trends in Testing; Implementation and Software Maintenance; System Control And Audit Trails; System Administration And Training; Hardware/Software Selection-Suppliers, Software Industry, Procedure and Phases in Selection of Software, Evaluation Process; Project Scheduling and Management.	
Text Books/References		
1.	R. Mall, Fundamentals of Software Engineering, PHI Learning , 2014	
2.	K. K. Aggarwal & Yogesh Singh, “Software Engineering”, 2ndEd., New Age International. 2005	
3.	R. S. Pressman, “Software Engineering – A practitioner’s approach”, 3rded., McGraw Hill Int. Ed.. 1992	
4.	Sommerville, “Software Engineering”, Pearson Education , 2006	

Central University of Kashmir

Course Title	Computer Networks	Course Code	BTCS 602	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	04	L	T	P
Semester	6th	4	0	0

Course Objectives

- To understand different components involved in transmitting data, data representation, and the direction of data flow.
- To develop an understanding of factors affecting the Network performance and to understand the basics of data, signals, transmission, and transmission media, Circuit switching and TDM, along with a brief overview of the telephone network.
- To understand the types of errors that can occur during data transmission, framing, and error detection and correction methods.
- To understand how different devices and protocols work together to enable communication between different hosts and networks in an internetworked environment. This knowledge is crucial for designing, implementing, and troubleshooting networks of various sizes and complexities.
- To understand how different layers and protocols work together to enable secure and efficient communication between different systems and applications over the network. This knowledge is crucial for designing, implementing, and managing secure and reliable networks.

Learning Outcomes

- Apply the knowledge in Selecting appropriate transmission media for a given application.

- Apply the knowledge in various networking scenarios, including designing and maintaining LAN, MAN, and WAN networks.
- Apply the knowledge of internetworking devices, addressing, routing techniques, and transport layer protocols to design and manage efficient and secure communication networks with improved quality of service.
- Apply the knowledge gained one can design and develop secure and efficient network applications and implement security measures to protect sensitive data and ensure secure communication.

Course Synopsis

The course aims to familiarize the students with all aspects of computer networking, from the basics of data communications and network criteria to advanced topics such as internetworking, routing protocols, and security, including a detailed study of networking layers and protocols such as TCP/IP, DNS, SMTP, SNMP, FTP, HTTP & WWW, and cryptography-based security measures like public and private key encryption and digital signatures.

Course Outline / Content

Unit	Topics	Weeks
1.	Overview of Networking: Introduction; Data communications: components, data representation (ASCII,ISO etc.), direction of data flow (simplex, half duplex, full duplex);network criteria, physical structure (type of connection, topology), categories of network (LAN,MAN,WAN); Internet: brief history, Protocols and standards; Reference models: OSI reference model, TCP/IP reference model, their comparative study, Delays, loss & throughput. Physical Layer fundamentals	4
2.	Data link Layer: Types of errors, framing(character and bit stuffing), error detection & correction methods; Flow control; Protocols: Stop & wait ARQ, Go Back- N, ARQ, Selective repeat ARQ,	4

	HDLC; Medium Access sub layer: Point to Point Protocol, LCP, NCP, Token Ring; Reservation, Polling, Multiple access protocols: Pure ALOHA, Slotted ALOHA, CSMA, CSMA/CD, CSMA/CA Traditional Ethernet, fast Ethernet(in brief)	
3.	Network layer: Internetworking & devices: Repeaters, Hubs, Bridges, Switches, Router, Gateway; Addressing : IP addressing, sub netting ; Routing : techniques, static vs. dynamic routing , Unicast Routing Protocols: RIP, OSPF, BGP; Other Protocols: ARP, IP, ICMP, IPV6; Transport layer: TCP&UDP, Process to Process delivery; Congestion Control: Quality of service: techniques to improve QOS : Leaky bucket algorithm, Token bucket algorithm	3
4.	Presentation & Session layer Description, functions, protocols and working. Application Layer Introduction to DNS, SMTP, SNMP, FTP, HTTP & WWW; Security: Cryptography (Public, Private Key based), Digital Signature, Firewalls.	4

Text Books	
1.	Computer Networks – Top Down Approach by Kurose & Ross, Pearson
2.	Data Communication & Networking by Forouzan, Tata McGraw Hill
3.	Computer Network, 4e, by Andrew S. Tenenbaum, Pearson Education/ PHI.
References	
1	Computer Networking, by Stanford H. Rowe, Marsha L. Schuh
2	Keizer, — Local Area Networks, McGraw Hill

3	Peterson and Davie, "Computer Networks, A Systems Approach", 5th ed., Elsevier, 2011
4	William Stallings: Data & Computer Communications, PHI

Central University of Kashmir				
Course Title	Theory of Computation	Course Code	BTCS 604	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	04	L	T	P
Semester	6th	4	0	0
Course Objectives				
• To understand basic kinds of finite automata and their capabilities. • To understand the use and applicability regular and context-free languages. • To understand the time and space complexity for p and np problems. • Constructing the Turing machine for Recursive languages.				
Learning Outcomes				
• Apply knowledge of FSMs to model control systems that have a finite number of states and inputs, such as traffic lights, vending machines and digital circuits. • Apply Regular expressions for searching and manipulating text data, and in text editors, scripting languages, and database systems. • Apply CFGs to define the syntax of programming languages in Compiler Design. • Apply the knowledge to identify the limits of computation, and provide a framework for understanding the complexity of algorithms and problems in computer science.				
Course Synopsis				
• Students can understand the concepts of automata, formal languages, grammar, computability and decidability				

- Students can understand how complex machine that we call a computer can be represented in an elegant and simple view.
- Students can understand how Automata Theory possesses a high degree of permanence and stability, in contrast with the ever-changing paradigms of the technology, development, and management of computer systems.
- Student can understand how Automata theory has direct bearing on practice:
 - o Such as on circuit design, compiler design.
 - o On cryptography and optimization problems.

Course Outline / Content

Unit	Topics	Weeks
1.	Basic Concepts: Symbols, Strings, Language, Formal Language, Natural Language. Basic Machine and Finite State Machine. FSM without output: Definition and Construction- DFA, NFA, NFA with epsilon Moves, Minimization Of FA, Equivalence of NFA and DFA, Conversion of NFA with epsilon moves to NFA, Conversion of NFA With epsilon moves to DFA. FSM with output: Definition and Construction of Moore and Mealy Machines, Inter- conversion between Moore and Mealy Machines. Regular Expressions: Definition and Identities of Regular Expressions, Construction of Regular Expression of the given L, Construction of Language from the RE, Construction of FA from the given RE using direct method, Conversion of FA to RE using Arden's Theorem, Pumping Lemma for RL, Closure properties of RLs, Applications of Regular Expressions.	4
2.	Introduction, Formal Definition of Grammar, Notations, Derivation Process: Leftmost Derivation, Rightmost Derivation, derivation trees, Context Free Languages, Ambiguous CFG, Removal of ambiguity, Simplification of CFG, Normal Forms, Chomsky Hierarchy, Regular grammar, equivalence of RG (LRG and RLG) and FA. Push Down	4

	Automata: Introduction and Definition of PDA, Construction (Pictorial/ Transition diagram) of PDA, Instantaneous Description and ACCEPTANCE of CFL by empty stack and final state, Deterministic PDA vs Non deterministic PDA, Closure properties of CFLs, pumping lemma for CFL.	
3.	Formal definition of a Turing machine, Recursive Languages and Recursively Enumerable Languages, Design of Turing machines, Variants of Turing Machines: Multi- tape Turing machines, Universal Turing Machine, Nondeterministic Turing machines. Comparisons of all automata.	4
4.	Decidability: Decidable problems concerning regular languages, Decidable problems concerning context- free languages, Un- decidability, Halting Problem of TM. Time Complexity: Measuring Complexity, The Class P, The Class NP, Examples of problems in NP, NP- completeness.	3

Text Books

1.	Michael Sipser, Introduction to the Theory of Computation, CENGAGE Learning, 3rd Edition ISBN: 978-0-321-2529-63.
2.	HopcroftUlman, Introduction to Automata Theory, Languages and Computations, Pearson Education Asia, 2nd Edition, ISBN: 9788131720479. 8458-7.

References

1.	Vivek Kulkarni, Theory of Computation, Oxford University Press, ISBN-13: 978-0 19-80
2.	Daniel Cohen, Introduction to Computer Theory, Wiley India, ISBN: 9788126513345, 2ed
3.	Kavi Mahesh, Theory of Computation: A Problem Solving Approach, Wiley India, ISBN: 9788126533114 9. BasavarajS.Anami, Karibasappa K.G, Formal Languages and Automata Theory, Wiley India, ISBN: 9788126520107

Central University of Kashmir				
Course Title	Computer Networks Lab	Course Code	BTCS 606L	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	02	L	T	P
Semester	6th	0	0	0
Course Objectives				
• To gain the firm understanding of networking concepts learned in the course work by Practical implementation of the same.				
Learning Outcomes				
• Students should understand the basics of computer networks, including the different types of networks and the components that make up a network. They should also be able to crimp network cables and wire them correctly. • Students should be able to use networking commands and tools to troubleshoot network issues, monitor network traffic, and configure network devices. • Students should understand the principles of firewall and VPN technologies, including how they protect networks from unauthorized access and how they enable secure remote access.				
Course Synopsis				
The objective of the lab course to familiarize students with networking concepts from a Practical perspective				
Course Outline / Content				
Unit	Topics			Weeks
1.	Introduction & Network Wire Crimping			1
2.	Networking Commands & Administration			1

3.	Ethernet & Token Ring	1
4.	Switched LANs	1
5.	Network Design	1
6.	ATM	1
7.	RIP: Routing Information Protocol	1
8.	OSPF: Open Shortest Path First	1
9.	TCP: Transmission Control Protocol	1
10.	Queuing Disciplines	1
11.	RSVP: Resource Reservation Protocol	1
12.	Firewalls and VPN	1
13.	Applications	1
14.	Project	1
Text Books		
1.	Computer Networks – Top Down Approach by Kurose & Ross, Pearson	
2.	Data Communication & Networking by Forouzan, Tata McGraw Hill	
References		
1.	Computer Networking, by Stanford H. Rowe, Marsha L. Schuh	
2.	Keizer, — Local Area Networks, McGraw Hill	
3.	Peterson and Davie, "Computer Networks, A Systems Approach", 5th ed., Elsevier, 2011	
4.	William Stallings: Data & Computer Communications, PHI	

Note: These are only the suggested list of practical. Instructor may frame additional practical relevant to the course content.

Central University of Kashmir				
Course Title	Microprocessor	Course Code	BTCS 607	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory	Department	Information Technology	
Credits	04	L	T	P
Semester	6th	4	0	0
Course Objectives				
• To acquire knowledge about the hardware architectures and the functional blocks of each microprocessors (8085 & 8086) • To know the functionality of common peripheral controllers and its interfaces with various peripheral devices • To gain the practical development of applications using microprocessors (8085 and 8086)				
Learning Outcomes				
• Technical knowhow in identifying the basic components of microprocessors • Ability to build a microprocessor/ microcontroller based system for practical applications • Understand the architecture of 8085 and 8086 • Impart the knowledge about the instruction set of 8085 and 8086 • Understand and apply the fundamentals of assembly language • Understand the basic idea about the data transfer schemes and its applications				
Course Synopsis				
Microprocessors course is intended to introduce the architecture, programming of microprocessors and interfacing various hardware circuits to microprocessors. This course analyze the complete architectural, programming, interfacing details of 8085 and 8086 microprocessors. Students completing this				

course will work with microprocessor based equipment, and be capable of distinguishing hardware from software faults. The superior students will also be capable of participating in product development efforts, including support and development of assembly language code.

Course Outline / Content

Unit	Topics	Weeks
1.	Introduction to microprocessor, Evolution and Applications of microprocessors. 8085 microprocessor: Architecture, Block diagram, Pin diagram of 8085 microprocessor.	4
2.	The 8086 Microprocessor Architecture: Introduction to 8086 microprocessor, Internal architecture of 8086, Memory segmentation, 8086 flags, Internal registers, physical address generation, Pin description in maximum & minimum modes.	4
3.	Addressing modes of 8086, Instruction execution timing, assembler instruction format, data transfer instructions, arithmetic instructions, branch instructions, looping instructions, NOP and HLT instructions, flag manipulation instructions, logical instructions, shift and rotate instructions, directives and operators, programming examples	4
4.	Interrupt definition, Maskable and non-maskable interrupts, vectored interrupts, 8086 interrupts, hardware, special instruction and condition produced by instruction Interrupt types, dedicated interrupts and software interrupts, Interrupt acknowledgement cycle, Interrupt priorities. I/O interfacing, Isolated I/O & memory mapped I/O, Memory interfacing.	4

Text Books/ References

1.	Microprocessor Architecture, Programming & Applications with 8085: Ramesh S Gaonkar; Wiley Eastern Ltd.
2.	Microprocessors and interfacing: Hall; TMH
3.	The Intel Microprocessors 8086- Pentium processor: Brey; PHI

4.	The 8088 & 8086 Microprocessors-Programming, interfacing, Hardware & Applications: Triebel & Singh; PHI
5.	Microcomputer systems: the 8086/8088 Family: architecture, Programming & Design: YuChang Liu & Glenn A Gibson; PHI
6.	Advanced Microprocessors and Interfacing: Badri Ram; TMH

Central University of Kashmir				
Course Title	Microprocessor Lab	Course Code	BTCS 608 L	
Degree	Bachelor of Technology	Branch	Computer Science & Engg.	
Course Type	Theory/ Laboratory	Department	Information Technology	
Credits	02	L	T	P
Semester	6th	0	0	2
Course Objectives				
• Understand the basic programming of microprocessor • Implement various operations on microprocessor kits • Design and implementation of interfacing • Design and implementation A/D Module • Design and implementation a stepper motor module				
Learning Outcomes				
• Obtain knowledge to do programs in assembly language programming using the trainer kits • Utilize development kits effectively for the real time applications of various peripheral devices with the processor • Hands on experience in interfacing devices such as A/D converter, D/A converter.				
Course Synopsis				
This is a two credit lab course which comes as hands-on experience on topics that are theoretically covered in the microprocessor course. The students write code in assembly language to develop a strong understanding of microprocessor fundamental operations. During this lab course, the student utilizes a real 8085 and 8086 microprocessor, different types of application, ranging from sensing simple environment parameter such temperature to controlling simple systems.				

Course Outline / Content		
Lab		Weeks
	1. Introduction to Microprocessor Kits and overview of Programming 2. Programs using basic programming construct 3. Program which used various control statements 4. Programs for interfacing 5. Programs on Timing and Delays 6. Interfacing of A/D Module 7. Interfacing of a stepper motor module	12
Text Books		
1.	Microprocessor Architecture, Programming & Applications with 8085: Ramesh S Gaonkar; Wiley Eastern Ltd.	
2.	Microprocessors and interfacing: Hall; TMH	
3.	The Intel Microprocessors 8086- Pentium processor: Brey; PHI	
4.	The 8088 & 8086 Microprocessors-Programming, interfacing, Hardware & Applications: Triebel & Singh; PHI	
5.	Microcomputer systems: the 8086/8088 Family: architecture, Programming & Design: YuChang Liu & Glenn A Gibson; PHI	
6.	Advanced Microprocessors and Interfacing: Badri Ram; TMH	